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Complexity Adjusted Comparisons of LPs and VARs: Evidence from Monte-Carlo Simulations

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in Evaluating Econometric Methods Using Monte Carlo Simulation

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1. Introduction

To infer statements about the dynamic response to exogenous shocks, the estimation of the corresponding impulse response has relied on two fundamental approaches, either through vector autoregressive (VAR) or Local Projection (LP) models. Since the introduction of the latter, an extensive literature has emerged discussing the relative merits of the two. A first strand of literature has established that, under relatively general conditions, LPs and VARs estimate the same population impulse responses (Plagborg-Møller & Wolf, 2021; Fusari et al. 2026). However, in the discussion of finite sample behavior, LPs were often viewed as less sensitive to dynamic misspecification whereas VARs were seen as potentially more statistically efficient (Li et al., 2024; Montiel Olea et al., 2025). Ludwig (2026) offers an analytical approach of explaining the observed finite sample bias-variance trade-off. He poses that LPs can be represented as a sequence of VAR estimations increasing in lag length as the estimated horizon increases and thus argues for a renewed thinking about how we can compare LPs and VARs – not by lag length, but by effective complexity.

The first part of this paper aims to build on the discussion outlined above by asking whether conventional finite-sample conclusions are robust to alternative benchmark comparisons that relax the standard equal-lag restriction. To answer this question, the paper operationalizes Ludwig’s (2026) central implication through a complexity-adjusted benchmarking exercise in which a fixed LP estimator is compared to progressively higher-order VAR specifications that proxy increasing effective dynamic flexibility (De Graeve & Westermarck, 2025). This comparison is conducted across empirically relevant sample sizes (see e.g. Herbst & Johannsen 2021) to assess how the resulting bias-variance trade-off depends on the available data.

The central research question is: Do LPs retain their apparent robustness advantage once they are compared not only to equal-lag VARs, but also to higher-order VARs that approximate the effective complexity implied by LP estimation? The contribution is not to establish a new LP-VAR equivalence result, but to ask whether standard Monte Carlo comparisons between LPs and equal-lag VARs are affected by the fact that LPs correspond to a more flexible dynamic benchmark.

The second part of the paper introduces controlled dynamic misspecification. While the baseline experiment studies complexity-adjusted LP-VAR comparisons under correctly specified linear dynamics, the main practical argument for LPs concerns their robustness when the estimated linear model is only an approximation to the true DGP (Montiel Olea et al., 2021). To study this channel explicitly, the paper introduces a controlled degree of misspecification into the DGP and examines how the relative performance of LPs, equal-lag VARs, and higher-order VARs changes as misspecification becomes more severe.

The simulation design proceeds in three steps. First, we consider a correctly specified bivariate VAR(4) DGP, allowing LP(4), VAR(4), and higher-order VARs to be compared under common autoregressive information. Second, we introduce dynamic misspecification by adding a moving-average shock component, thereby generating VARMA(4,1) dynamics while continuing to estimate finite-order VARs and LPs. Third, we introduce nonlinear misspecification through quadratic terms in the DGP while retaining linear estimators. This structure separates the role of dynamic approximation error from nonlinear functional-form misspecification and allows us to study whether conventional LP-VAR finite-sample rankings are robust to complexity-adjusted VAR benchmarks.

For each estimator, horizon, DGP, and sample size, we compute bias, variance, mean squared

error, and empirical coverage rates. Bias and variance describe the finite-sample trade-off in point estimation, while coverage assesses whether the corresponding inference procedures re-main reliable under increasing dynamic or nonlinear misspecification.

2. Theoretical Background

In this section, we will establish the theoretical background for our Monte Carlo simulations by discussing the relevant literature on Vector Autoregressions (VARs) and Local Projections (LPs). Further, we will discuss the theoretical results on the equivalence of VARs and LPs in terms of population and in sample impulse responses.

2.1 Introduction to VARs

Here I will write an introduction to VARs, their estimation and inference, and the corresponding impulse response functions. I will also discuss the potential issues with VAR estimation, such as lag length selection and dynamic misspecification.

2.2 Introduction to Local Projections

Here I will introduce the concept of Local Projections (LPs) as an alternative to VARs for estimating impulse responses. I will explain how LPs work, their advantages and disadvantages compared to VARs, and the conditions under which they are equivalent to VARs in terms of population impulse responses.

2.3 On the Equivalence of VARs and Local Projections

Here I will discuss the theoretical results that establish the equivalence of VARs and LPs under certain conditions, as shown by Plagborg-Møller & Wolf (2021) and Fusari et al. (2026). I will also explain how this equivalence is affected by finite sample properties and dynamic misspecification, as discussed by Li et al. (2024), Montiel Olea et al. (2025), and Ludwig (2026).

The conventional finite-sample comparison between local projections and vector autoregressions is typically conducted at equal lag order, pitting $LP(p)$ against $VAR(p)$. Ludwig (2026) shows that this convention rests on a misleading notion of parsimony. His Theorem 1 establishes an exact finite-sample identity: the h -step prediction of an $LP(p)$ is the forward iteration of a sequence of one-step VAR predictions whose lag order rises from p to $p + h - 1$. An equal-order $VAR(p)$ is therefore a restricted special case of $LP(p)$, obtained by holding the lag order fixed along the recursion rather than allowing it to increase. The familiar empirical pattern—lower approximation bias but higher sampling variance for LP relative to a same-order VAR—is, from this vantage point, largely a mechanical consequence of comparing a more general specification with its own restriction, rather than evidence of an intrinsic property of either estimator. This matters because it leaves the standard reading of the bias-variance trade-off, as articulated by Plagborg-Møller and Wolf (2021) and Montiel Olea et al. (2025), confounded: at equal lag order, the method effect (LP versus VAR) cannot be separated from the complexity effect (more versus fewer effective autoregressive parameters).

3. Baseline Simulation

This confounding motivates the first part of our study. Because Ludwig’s contribution is analytical and deliberately abstains from a simulation-based horse race, the question of whether the conventional finite-sample conclusions survive once the equal-lag restriction is relaxed remains open. We therefore operationalize Ludwig’s central implication through a complexity-adjusted benchmarking exercise that holds parsimony more nearly fixed, comparing a fixed LP(4) estimator against progressively higher-order VAR specifications that proxy its increasing effective dynamic flexibility. Crucially, because the identity in Theorem 1 maps LP(p) to a sequence of VARs rather than to any single higher-order VAR, no individual VAR is the appropriate counterpart; instead, LP(4) is read as a reference line against the full VAR order frontier, against which its bias, variance, and coverage can be located. We frame the exercise as a test of robustness rather than as a foregone conclusion: the aim is to assess whether the conventional bias-variance verdict persists, dissolves, or reverses once the comparison is placed on more comparable terms, and to identify the features of the data-generating process under which each procedure is favored.

3.1 Simulation Design

3.2 Simulation Results

4. Extension 1: Dynamic Misspecification

4.1 Simulation Design

4.2 Simulation Results

5. Extension 2: Functional Misspecification

5.1 Simulation Design

5.2 Simulation Results

6. Discussion

Literatur

Ludwig, J. F. (2026). Local projections are VAR predictions of increasing order. *Available at SSRN 6441981*.

Declaration of independence